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Scope of Solar Energy and Wind Energy in District Raebareli (Uttar Pradesh)

Harshit Pratap Singh

Abstract: ***Renewable energy resources are the Potentials for today and for the forthcoming future of our Generations thus making the relieve from dependability of generations on the fossil fuels and making the right use of resources of what nature has given to us . Talking about the todays existing renewable energies are in the form of Solar, Wind, Tidal, Hydrothermal, Geothermal etc are various kind of sources of energies in the Renewable forms. Talking about the scope of innovations in this area it has shown a great potentials as we can know about the Technology of Decarbonisation in which carbon is been seeing as source of energy Potentials for the future. Talking about the todays worlds consumption , it is about Coal(27%) , Natural gas (24%) , Hydro energy (7%). Talking about Raebareli district in Uttar Pradesh the average Specific Photovoltaic Power Output is found to be 1495.6 kWh/kWp , Direct Normal Irradiation found to be 1190.9 kWh/m² and Global Horizontal irradiation is found 1771.5kWh /m².So according to these datas it can be conferred that a great scope of solar energy can be utilised in my area and also the Government has some programmes also for installing the solar Panels in thus area .***

Keywords: ***Solar Irradiance, Wind Power, Solar Power Solar Energy***

I. INTRODUCTION

The Solar energy is a very boon for us and also they can be used in various fields of life by reducing the reliability of us from the Non renewable sources of energy as they are very limited and are depleting very fast. So in the Raebareli district as its located on the Plains regions so amount of wind is not much of strength but at the same time the amount of solar energy receiving per yearly are much of that capacity that it can be utilised in the forms of solar power for the Households Lightining not only but for all the Purposes likewise in agricultural fields also. So, the data obtained is from the Website of Nasa Larc .We have Collected the data from that site and does analysis of wind and solar power for the future Installement of any projects In Raebareli.

II. ANALYSIS

3.1 Wind Power

Average speed of wind :- 5.005m/s
Maximum Speed of Wind : 9 m/s
Minimum Speed of wind : 2.01 m/s

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* Correspondence Author
Harshit Pratap Singh*, Pursuing Master, Department of Electrical Engineering, Chandigarh University, Punjab, India.

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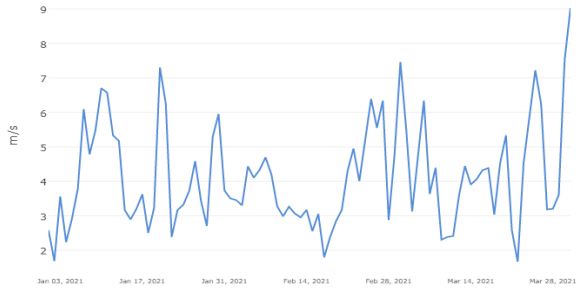

Figure 1.1

3.1.1 Outcome:

So from the above Data we can easily observed that amount of wind been blowing is just below the minimum wind required to generate the Wind energy (6.42m/s minimum wind speed). So, by yearly as the Wind is not same whole throughout the year hence its not possible to install wind power in Raebareli.

3.2 Solar Irradiance

Average Specific Photovoltaic Power Output: 1495.6 kWh/kWp
Direct Normal Irradiation :1190.9 kWh/m²
Global Horizontal irradiation :1771.5kWh /m²

3.2.1Optimum Tilt of Solar Panels by Month

January	February	March	April	May	June
34 °	26 °	18 °	10 °	2 °	180 °
July	August	September	October	November	December
2 °	10 °	18 °	26 °	34 °	42 °

Figure 1.2

3.2.2 Outcome:

So from the above data we can see that the Average amount of solar energy receiving is very much efficient to Generate the Solar Power hence there is a great scope of Using Solar Energy in Rae bareli and Not Only that these Solar Power Can be Used to Generate Power to Run motor Powers Like Tube wells for the Agricultural Fields and thus it have a very large Diverse of Applications.

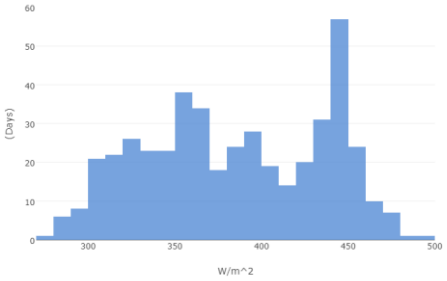

Figure 1.3

III. CONCLUSION

So we can conclude from the Above data that the Rae bareli has a great Potentials and future for the Solar Based Power Plants Installment and also these solar Power can be Utilised in various forms of Applications . Also the Availability of Wind Power is not so much Efficient and hence Wind Energy has a very negligible scope o f existence.

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AUTHORS PROFILE

Harshit Pratap Singh, Master's in Electrical engineering, Bachelor's in Electrical Engineering, Chandigarh University. Currently Pursuing Master's in Electrical Engineering. I am Determined to take up my Research Aptitude in the Field of Energy Resources basically Dealing with Renwable Energy and Resources. It is my First Paper to be Published and I am going to consistently send numerous of my Papers in this Field of Research. I am Very Passionate about exploring the ways of Renwable Energy and my Next Publication will be in the Fields of Power system and the Role of Renewable Energy in the Power system.

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